The Invariance Principle — problem 1.2

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Start with the set $\{3, 4, 12\}$. In each step you may choose two of the numbers a, b and replace them by $0.6a - 0.8b$ and $0.8a + 0.6b$. Can you reach the goal (a) or (b) in finitely many steps :
(a) $\{4, 6, 12\}$, (b) $\{x, y, z\}$ with $|x - 4|, |y - 6|, |z - 12|$ each less than $1/\sqrt{3}$?